Medical Diagnosis and Report Generation for Endoscopic Procedures Using Vision-Language Models

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B.Tech CSE Semester 8 Final Evaluation

Agenda

1. Introduction
2. Methodology
3. Architecture
4. Implementation
5. Results
6. Future Work
7. Conclusion

Introduction

Motivation & Problem Statement

The Clinical Burden:

- Each endoscopic procedure requires a detailed written report
- Manual report writing: **3–5 minutes per image** — a major bottleneck in busy GI units
- Documentation fatigue contributes to physician burnout Eaton [2024], Hurst [2024], National Academies of Sciences, Engineering, and Medicine [2024]

The AI Gap:

- General-purpose MLLMs (GPT-4V, LLaVA) lack specialised endoscopy terminology and pathological reasoning
- No model trained on physician-annotated GMC Calicut endoscopic data existed

Our Objective:

- Develop **MVLMERG** — a Multimodal Vision-Language Model for Endoscopic Report Generation
- Two-stage P-LoRA fine-tuning on publicly available GI datasets + GMC Calicut physician annotations
- Target: **automated, structured, interpretable** endoscopic reports in $\sim$7 seconds

Problem Definition

Core Problem:

- Writing endoscopic image reports requires deep domain expertise in gastroenterological terminology, lesion morphology, and clinical logic
- General Multimodal Large Language Models (MLLMs) exhibit critical knowledge gaps: they struggle with specialised endoscopy terminology, pathological feature descriptions, and complex medical reasoning
- Existing fine-tuning methods suffer from **limited domain adaptability** and **insufficient endoscopic-specific knowledge**
- High-quality physician-annotated endoscopic data is scarce — publicly available medical datasets (PubMed) cover general medicine but lack the specificity needed for endoscopic report generation

Goal: Develop **MVLMERG** — a fine-tuned vision-language model that bridges the domain gap, producing professional-grade endoscopic reports matching the standard of intermediate-level physicians. Ali et al. [2024], Haghighi et al. [2024]

Related Work

General Medical MLLMs:

- **LLaVA-Med** Li et al. [2023] — trained on PMC-15M biomedical image-text pairs; strong generalization but lacks endoscopy specificity
- **HuatuoGPT-Vision** — incorporates medical knowledge but insufficient for endoscopic detail
- **BiomedGPT, Uni-Med** — broad medical coverage; weak on GI endoscopic scenarios

Surgical / GI-specific models:

- **SurgicalGPT** — 3–5% accuracy gain in surgical VQA
- **ColonGPT / ColonINST** — colonoscopy instruction tuning

Fine-Tuning Methods:

- **LoRA** Hu et al. [2022] — low-rank weight adaptation; efficient but no domain-specific prompt alignment
- **Prefix / P-Tuning** Liu et al. [2022] — soft prompt tokens; needs combination with weight adaptation
- **PeFoMed, MISS** — parameter-efficient medical MLLM fine-tuning; not endoscopy-specific

Gap MVLMERG fills:

- Combines **P-LoRA** (prompt tokens + selective LoRA) with GMC Calicut physician-annotated data
- First model targeting professional Indian

Methodology

Dataset Creation and Augmentation

- **Data Source:** Government Medical College (GMC), Kozhikode
- **Raw Data:** 6,500 **patient reports** (semi-structured PDFs + endoscopic images)
  - Each report includes **4 endoscopic images**
- **Data Augmentation:**
  - Applied **rotation and random translation** to every image
  - Purpose: increase dataset size and improve model robustness
  - **Approximate scale:**
    - 6,500 reports $\times$ 4 images $=$ 26,000 base images
    - Rotation alone multiplies by 4 $\rightarrow$ over 100,000 images
    - Translation variations further expand the pool
- **Text Data:**
  - Final Q&A pairs derived from report content: **32,545 unique samples**
  - Data split: 30,917 training / 1,627 validation
- **Multimodal Association:** Each Q&A pair linked to original + augmented endoscopic images Wang et al. [2024], Zhang and et al. [2024]

Figure 1: Custom Dataset Workflow

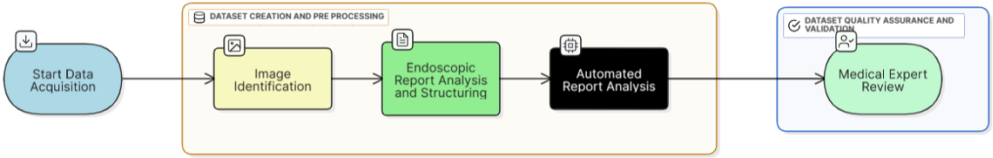


Figure: Data Workflow: From 6,500 patient reports to 32,545 Q&A pairs Wang et al. [2024], Zhang and et al. [2024]

Data Augmentation: Rotation

- Each original endoscopic image is rotated by **0°, 90°, 180°, and 270°**
- Produces exactly **4 variants** per image
- Clinically safe: tissue textures and lesion appearances remain unchanged
- **Effect:** $4\times$ structured expansion of the base image set

Data Augmentation: Stochastic Translation

- **Sampling:** Bernoulli process with $p = 0.2$ selects approx. 20% of rotated images for translation
- **Shift model:** Translation vector (τ_x, τ_y) drawn from a truncated Gaussian:

$$\tau_x \sim \mathcal{N}(0, (0.05\,W)^2), \text{ clamped to } [-0.15\,W, +0.15\,W]$$
$$\tau_y \sim \mathcal{N}(0, (0.05\,H)^2), \text{ clamped to } [-0.15\,H, +0.15\,H]$$

- **Overlap guarantee:**

$$\text{Overlap Fraction} = \left(1 - \frac{|\tau_x|}{W}\right) \times \left(1 - \frac{|\tau_y|}{H}\right) \geq (1 - 0.15)^2 = 72.25\%$$

- **Result:** Controlled positional variation mimicking endoscopic camera jitter, without harming diagnostic content.

Data Augmentation Examples

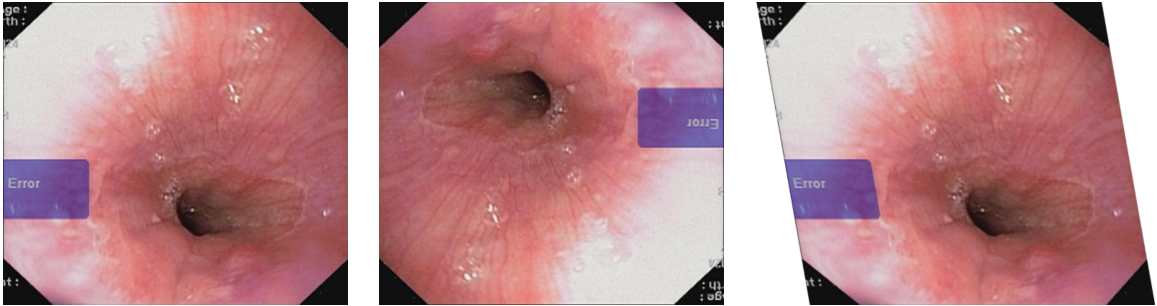
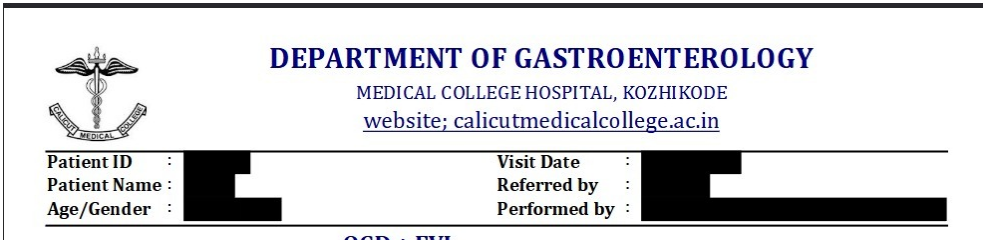


Figure: Left: Original Middle: Rotated (90°) Right: Randomly translated

Patient Data De-identification & Ethical Compliance

- **Regulatory Framework & Automated Script:**
  - Strict adherence to the **Digital Personal Data Protection (DPDP) Act**, supported by a formal MoU between GMC Kozhikode and NIT Calicut
  - **Redaction approach mirrors rigorous de-identification used in recent high-profile document releases (e.g., Epstein files)** – all personally identifiable fields irreversibly blacked out to ensure zero traceability
  - **Header Field Redaction:** Automatic blackout of Patient ID, Name, Age/Gender, Visit Date, UGI/OGD Numbers
- **Result:** Irreversible anonymization of all sensitive patient information before data enters the pipeline



DEPARTMENT OF GASTROENTEROLOGY

MEDICAL COLLEGE HOSPITAL, KOZHIKODE

website; calicutmedicalcollege.ac.in

Patient ID :	[redacted]	Visit Date :	[redacted]
Patient Name :	[redacted]	Referred by :	[redacted]
Age/Gender :	[redacted]	Performed by :	[redacted]

Report Analysis and Preprocessing

Medical Entities Extracted: Wang et al. [2024], Zhang and et al. [2024]

- Pathological findings and characteristics
- Diagnostic conclusions
- Procedural techniques and instruments

Processing Method:

- Natural language processing techniques
- Extraction of critical medical elements
- Foundation for Q&A pair generation

Automated Question-Answer Pair Generation

Multi-Stage Process: Hicks [2024], Pogorelov and et al. [2024]

1. **Medical Entity Extraction** – Key concepts and diagnoses
2. **Question Template Design** – Diagnostic, anatomical, pathological, procedural, comparative
3. **Automated QA Generation** – Multiple question variations per concept Liu et al. [2024], Li et al. [2023]
4. **Answer Synthesis** – Accurate answers from reports + medical knowledge

Result: Diverse, clinically relevant question-answer combinations.

Dataflow Pipeline and Methodology

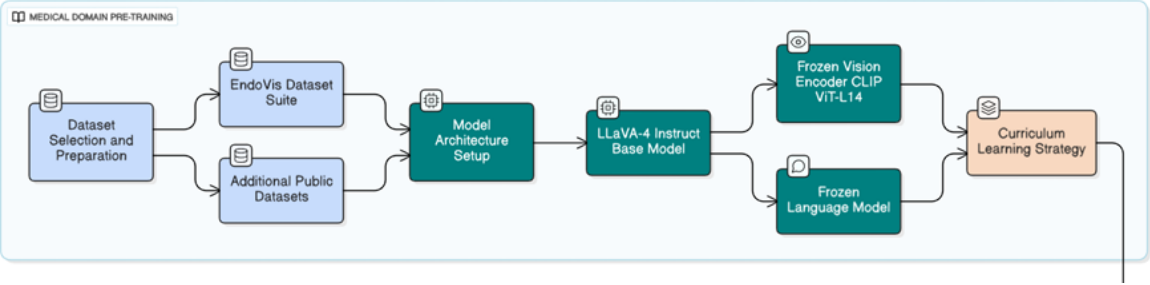


Figure: Stage 1: Pre-training.

Dataflow Pipeline and Methodology

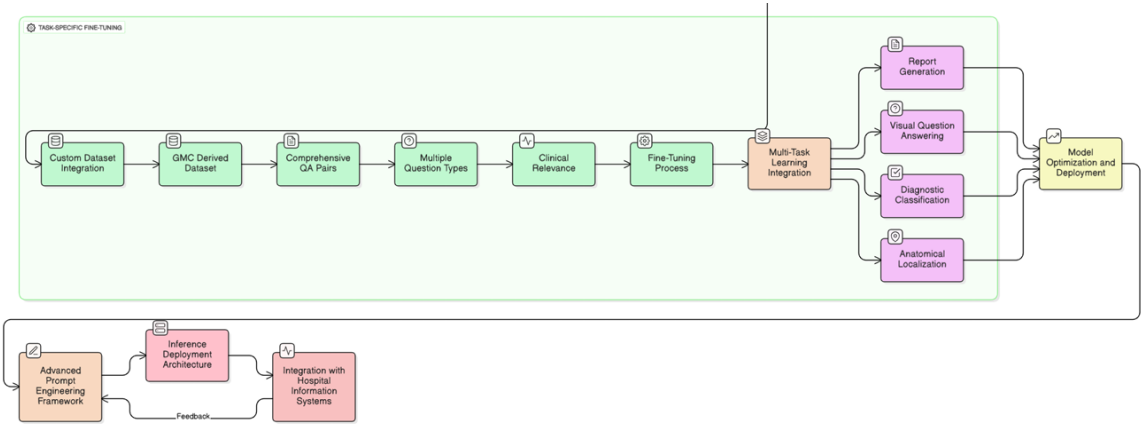


Figure: Stage 2: Fine-tuning

Architecture

Two-Stage Training Framework — MVLMERG

Stage 1: Medical Domain Knowledge Injection Leaman and Lu [2024], Hu et al. [2022]

- **Objective:** Acquire gastrointestinal domain knowledge
- **Base Model:** LLaVA-4 with frozen CLIP ViT-L/14
- **Trainable:** LLM backbone (vision encoder + connector frozen)
- **Datasets:** PubMed, MedTrinity-25M, VQA-Med ($\sim$60K GI records) Xie et al. [2025], Ben Abacha et al. [2019]

Stage 2: Professional Report Generation Leaman and Lu [2024], Hu et al. [2022]

- **Objective:** Learn physician-style endoscopic report language
- **Dataset:** GMC Calicut physician-annotated data (6,500 base; $\sim$31K after augmentation)
- **Strategy:** P-LoRA — prompt tokens + selective LoRA on key layers Hu et al. [2022], Liu et al. [2022]
- **Unfrozen:** LLM backbone (vision encoder + connector frozen)

P-LoRA: Prompt-based LoRA Fine-Tuning

Core Innovation of MVLMERG Hu et al. [2022], Liu et al. [2022]

Soft Prompt Tuning:

- Learnable tokens $p_1, \ldots, p_m$ prepended to input
- Guides model toward domain-specific language patterns
- No manual prompt crafting — adapts through training

Selective LoRA:

- Applied only to first, middle, and last layers
- $\Delta W = AB$, rank $r = 64$, $\alpha = 32$
- Only **0.343%** of parameters are trainable

Why P-LoRA beats LoRA alone?

- Prompt tuning aligns input representations
- LoRA adapts weight matrices efficiently
- Combined: +5% CIDEr, +4.4% ROUGE-L over plain LoRA
- GPU memory: ∼43 GB during training

Inference:

- ∼7s per report on A100 80GB
- Deployable on RTX 3090 (24 GB)

System Architecture

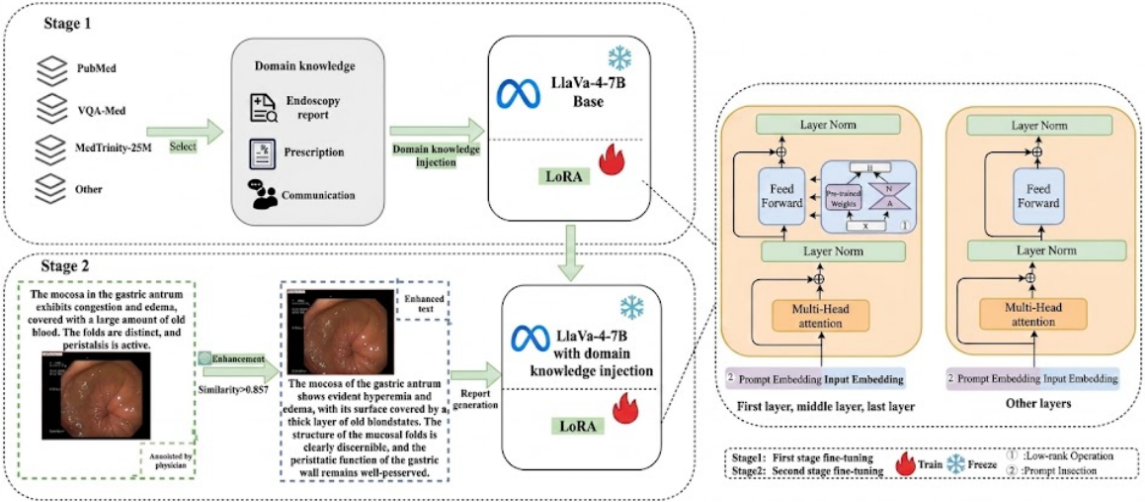


Figure: High-level architecture of the proposed vision-language model for endoscopic report generation.

Model Architecture – LLaVA

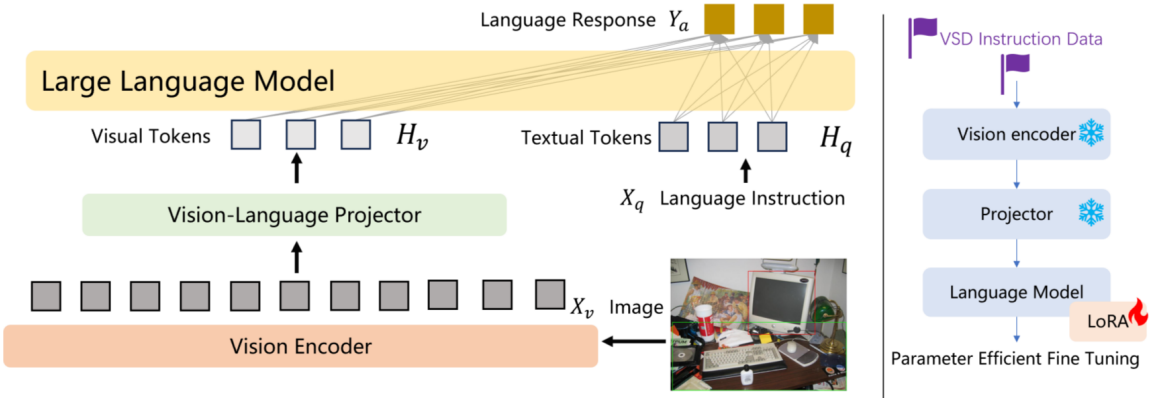


Figure: LLaVA architecture: vision encoder + projection layer + language model for multimodal understanding.

CLIP-ViT Architecture

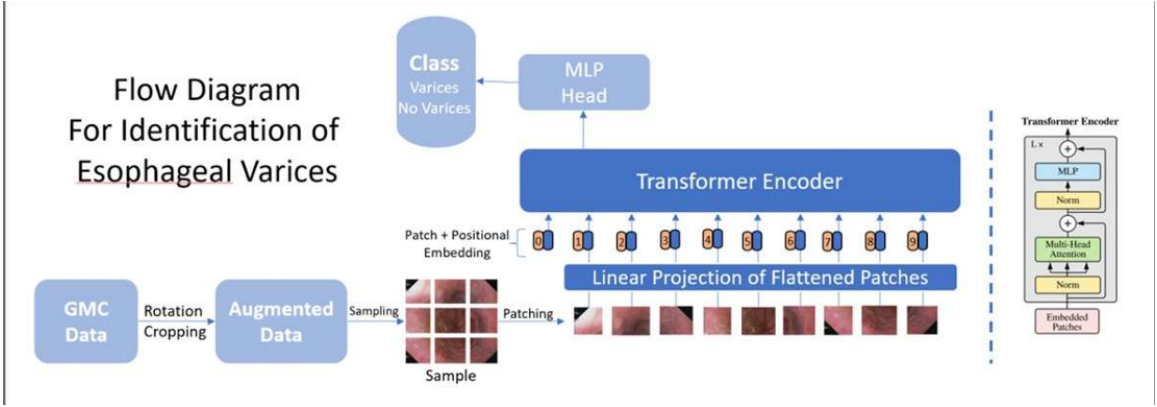
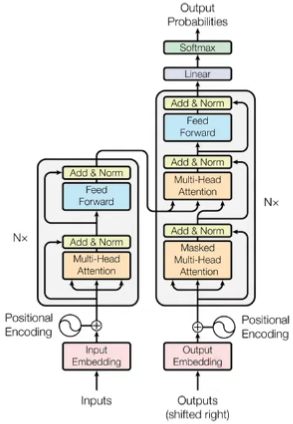
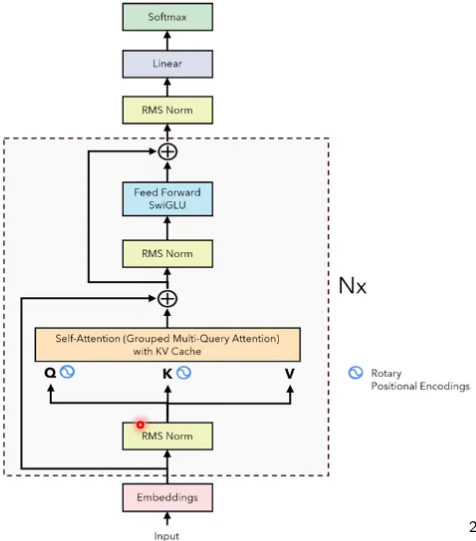


Figure: CLIP-ViT-L/14 vision encoder: patch embedding, transformer layers, and projected visual features.

Transformer vs LLaMA





Implementation

Refining Generation: Implementing System Prompts

Initial Challenge:

- Early model iterations produced verbose, unstructured, and redundant reports that did not match standard clinical formatting.

Implemented Solution (Dataset Reformatting):

- Integrated rigorous **System Prompts** directly into the training dataset.
- **Role Assignment:** Instructed the model to assume a specific clinical persona (e.g., "You are an expert gastroenterologist...").
- **Output Constraints:** Provided explicit instructions to prioritise direct, concise, and structured answers mirroring real GMC Kozhikode reports.

Result: A massive reduction in unnecessary text, yielding highly concise and structured outputs; further refinement improved diagnostic accuracy.

Qualitative Impact: Verbose vs. Concise Reports

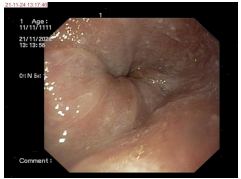
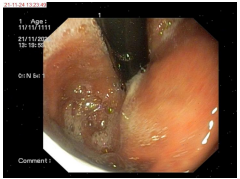
Previous Generation
(Verbose & Redundant)

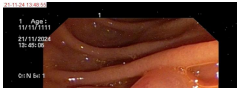
Medical Endoscopy Report

Patient Information

Patient Name: ABDUL MANAF
Patient ID: ABDUL MANAF_86376_RP241121_015818
Report Date: 2025-12-04 12:55:59

Endoscopy Images



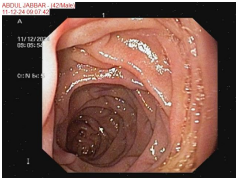
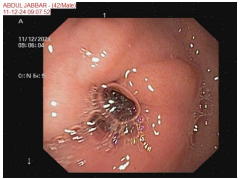
Current Generation
(Concise via System Prompt)

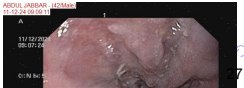
Medical Endoscopy Report

Patient Information

Patient Name: ABDUL JABBAR
Patient ID: ABDUL JABBAR_84105_RP241211_091205
Report Date: 2026-03-06 13:45:39

Endoscopy Images



Qualitative Impact: Verbose vs. Concise Reports

Previous Generation
(Verbose & Redundant)

Scope Passed Upto:	D2 mucosa without ulcers/ erosions/ luminal narrowing .Scope passed upto D3 mucosa without any difficulty .Biopsy taken from D2 and D3 mucosa for HPR and CBNAAT from D2 and D3 mucosa . D1 mucosa was edematous and erythematous with superficial ulcers . D1 was not entered further due to poor patient tolerability . D1 mucosa was edematous and erythemat
Premedication:	10% XYLOCAINE ANTRUM AND PYLORUS WITH SURROUNDING NODULARITY AND D1 NODULARITY FOR BIOPSY TAKEN RUT: PLAN: CECT THORAX + ABDOMEN AND BARIUM SWALLOW IF FURTHER EPISODE/HB FALL BIOPSY TAKEN FROM NODULARITY AND D1 NODULARITY FOR CBNAAT TAKEN RUT: PLAN: CECT THORAX + ABDOMEN AND BAR
Esophagus:	Normal mucosa without ulcers/ erosions/ luminal narrowing with normal peristalsis and distensibility noted .No foreign body seen in the esophagus .No esophageal varices seen .No altered blood seen in the lumen .No luminal blood seen in the esophagus .No foreign body seen in the esophagus .No altered blood seen in the lumen .No luminal blood seen in the esophagus .No foreign body seen in the esophagus .No altered blood seen in the lumen .No luminal blood seen
Ge Junction:	40CMS and edematous and erythematous mucosa noted.Biopsy taken .No nodularity or ulcerations noted. No tear noted at present. No salivary pooling. No difficulty in swallowing. No foreign body seen. No luminal blood seen. No active ooze seen. No hemoptysis. No melena. No hematemesis. No hematuria. No proteinuria. No renal papilla. No ureter. No ampulla. No pyloric opening. No duodenal ulcer. No
Stomach:	No stasis . Normal peristalsis and distensibility noted . No ulcer/ erosions noted . Normal blood in lumen . No active bleed at present . Normal mucosal folds and contours noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal
Fundus:	Normal mucosa without ulcers/ erosions/ luminal narrowing with normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal

Current Generation
(Concise via System Prompt)

Stomach:	
Fundus:	No varices
Body:	Mosaic pattern of mucosa seen
Antrum:	Normal
Pylorus:	Normal
Duodenum:	D1: Normal | D2: Normal
Rut:	Mosaic pattern of mucosa seen
Biopsy:	Not taken
Impression:	GRADE II X3 COLUMNS OF ESOPHAGEAL VARICES WITHOUT RCS SEVERE PHG

Report generated by Medical LLaVA AI on 2026-03-06 at 13:45:39

The “Blackbox” Problem & Our Solution

The Initial Blackbox:

- The model would output a final diagnosis directly from an image without revealing its reasoning, making it hard to trust or debug.

Our Solution – Chain-of-Thought Integration:

- Augmented the training Q&A pairs with **step-by-step reasoning chains**.
- Each chain forces the model to structure its answer as: **anatomical identification** $\rightarrow$ **visual observation** $\rightarrow$ **diagnosis**.
- During training, the model learns to mimic this transparent reasoning process.

Outcome: The model now explicitly explains *what it sees* and *why it concludes something*, turning a black-box predictor into an **interpretable diagnostic assistant**.

Chain-of-Thought (CoT) in Endoscopy

Before CoT Implementation (Baseline Model):

- **Input:** [Image] "What is the diagnosis?"
- **Output:** "Grade II Esophageal Varices." *(correct but no justification)*

After CoT Implementation (Our Model):

- **Input:** [Image] "Analyze the image step-by-step and provide a diagnosis."
- **Output:**
  1. **Observation 1:** I observe the lower third of the oesophagus.
  2. **Observation 2:** There are swollen, tortuous submucosal veins present.
  3. **Observation 3:** The veins occupy less than 1/3 of the lumen, with no red colour signs.
  4. **Conclusion:** Therefore, the diagnosis is Grade II Esophageal Varices.

Impact: Explicit reasoning enhances clinical trust, allows verification, and helps identify subtle mistakes – a crucial feature for real-world deployment.

Fine-Tuning Configuration Details

Vision Encoder: CLIP-ViT-L/14

- Image resolution: 336×336 pixels
- Patch size: 14×14 (24×24 grid)
- Total vision tokens: 577 per image (576 patches + CLS)

Training Hyperparameters:

- **Precision:** bfloat16 (BF16)
- **Optimizer:** AdamW with 8-bit quantization
- **Learning Rate:** $2e^{-4}$ with cosine scheduler, warmup 0.1
- **Batch Size:** 2 micro-batch, 4 accumulation steps = 8 global
- **Attention:** Flash Attention 2
- **Epochs:** 4 epochs (both stages)

Training Configuration Rationale

- **CLIP-ViT-L/14** – pre-trained on 400M pairs; robust medical visual features; L size balances capacity vs. speed; 14×14 patch grid captures both anatomy and small lesions.
- **336×336 resolution** – standard for LLaVA; sufficient detail without excessive memory.
- **577 vision tokens** – each patch token carries localised information, enabling the language model to "see" global and local structures.
- **bfloat16** – halves memory vs. FP32, no performance loss.
- **AdamW + 8-bit** – stable weight-decoupled convergence; 8-bit reduces optimizer memory 75%.
- **LR $2e^{-4}$, cosine + warmup** – low peak prevents catastrophic forgetting; warmup stabilises early steps.
- **Batch size 8 (accumulated)** – simulates larger batch under GPU memory limits.
- **Flash Attention 2** – memory-efficient exact attention; 5-10× VRAM reduction for long sequences.
- **4 epochs** – sufficient for P-LoRA adapters on the augmented second-stage dataset without overfitting.

Why AdamW & Flash Attention 2?

AdamW over Adam

- Adam couples weight decay with gradient scaling – can over-regularize.
- AdamW **decouples** weight decay from adaptive gradients $\rightarrow$ better generalization.
- Standard for LLaMA fine-tuning; prevents over-shrinking of LoRA adapters.

Flash Attention 2 over regular / Flash Attention 1

- Standard attention: $O(N^2)$ memory – prohibitive for 577 vision tokens + text.
- Flash Attention 1 tiled computation, but under-utilised GPU parallelism.
- Flash Attention 2:
  - **Fewer non-matmul operations** $\rightarrow$ higher throughput.
  - **Better batch/head parallelism** $\rightarrow$ faster training.
  - **Lower VRAM usage (5-10$\times$ reduction)** $\rightarrow$ feasible on single GPU.

Together: Stable convergence + memory-efficient training at scale.

Results

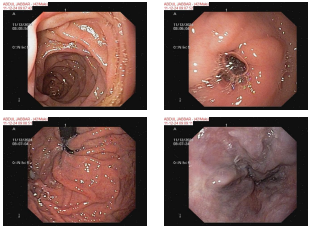
Generated Report

Medical Endoscopy Report

Patient Information

Patient Name:	ABDUL JABBAR
Patient ID:	ABDUL JABBAR_84105_RP241211_091205
Report Date:	2026-04-13 18:09:26

Endoscopy Images



Findings

Indication:	CLD-UGIB
Scope Passed Upto:	D2
Premedication:	Lignocaine spray
Esophagus:	Normal
Ge Junction:	Lax at 38cm
Stomach:	
Fundus:	Edematous mucosa noted
Body:	Edematous mucosa noted
Antrum:	Erythematous mucosa noted
Pylorus:	Normal
Duodenum:	D1: Normal | D2: Normal
Rut:	Not taken
Biopsy:	Not taken
Impression:	GRADE II ESOPHAGEAL VARICES WITH RCS EVL DONE SEVERE PHG

Report generated by Medical LLaVA AI (FFT) on 2026-04-13 at 18:09:26

Final diagnostic report – Pages 1 and 2

Quantitative Comparison — MVLMERG vs. Baselines

Method	CIDEr↑	ROUGE-L↑	METEOR↑	BLEU-4↑	BERTScore↑
Qwen-VL-7B	0.0212	0.2815	0.2369	0.2293	0.6947
VisualGLM-6B	0.0147	0.2210	0.1913	0.1829	0.5893
GPT-4V	0.0955	0.3040	0.3311	0.3292	0.7846
HuatuoGPT-Vision-7B	0.0802	0.2642	0.2791	0.2709	0.7538
LLaVA-Med	0.0171	0.1979	0.2677	0.2417	0.7452
BiomedGPT	0.0233	0.1739	0.2381	0.2178	0.7117
Uni-Med	0.0881	0.2013	0.1928	0.1903	0.6639
EndoChat	0.0578	0.1965	0.2102	0.1961	0.6740
MVLMERG (Ours)	**0.2667**	**0.4598**	**0.4070**	**0.3863**	**0.8216**

MVLMERG achieves state-of-the-art on all 5 metrics ($p < 0.01$, Cohen's $d > 0.8$). Report accuracy: **84.03%**.

Report Accuracy & Calibration

Expert Physician Evaluation Ali et al. [2024]

- 2 senior endoscopists (10+ years experience), GMC Calicut
- Reports labelled Correct/Incorrect; disagreements resolved by joint discussion

Method	Accuracy ↑
HuatuoGPT-Vision-7B	0.7240
LLaVA-Med	0.7013
BiomedGPT	0.6426
Qwen-VL-7B	0.5659
EndoChat	0.6734
GPT-4V	0.7833
MVLMERG (Ours)	**0.8403**

Calibration Analysis:

- ECE (Expected Calibration Error) = **0.042**
- Token-level confidence binned into 10 intervals
- Confidence aligns closely with actual correctness
- Model is **well-calibrated** — predictions are reliable, not overconfident

Inference Speed:

- ∼**7 s/report** on A100 80GB
- vs. 3–5 min manual writing
- **26**× **faster** than manual reporting

Hardware & Training Setup

Training Hardware:

- **GPU:** NVIDIA A100 80GB PCIe (CUDA 12.9)
- **CPU:** Intel Xeon Silver 4310 @ 2.10 GHz (24 physical cores, 48 threads, 2 sockets)
- **RAM:** 125 GiB
- **Python:** 3.10.12; PyTorch (BF16 + Flash Attention 2)

Training Resource Usage:

- GPU memory during training: $\sim$43 GB
- Stage 1 training time: $\sim$70 h
- Stage 2 training time: $\sim$15 h
- Trainable parameters: **0.343%** of full model

P-LoRA Hyperparameters:

- LoRA rank $r = 64$, $\alpha = 32$, dropout $= 0.05$
- Prompt tokens $= 64$
- Epochs: 4 (both stages)
- Seed: 42
- Max sequence length: 4096

Minimum Deployment Spec:

- 16 GB RAM, Python 3.8+
- CUDA 11.2+, PyTorch 2.0+
- Can run on RTX 3090 (24 GB)
- $\sim$7 s inference latency

Future Work

Model Optimization & Quantization

GGUF Format Conversion Wong [2024], SandLogic Technologies [2024]

- Optimizes CPU and mixed CPU-GPU inference
- Significantly improved model loading times Corp. [2024], Wong [2024]

Multi-Level Quantization Strategy SandLogic Technologies [2024], Gupta [2024]

Format	Use Case	Size Reduction
Q8_0	High-end workstations	∼50%
Q5_K_M	Standard clinical computers	∼69%
Q4_K_M	Basic clinical terminals	∼75% (3.25GB) TheBloke [2024]
Q3_K_L	Mobile deployments	Ultra-compact

Deployment Planning

Two Deployment Strategies:

1. **vLLM Framework** Fregly [2024], Linh [2024]
   - High-throughput clinical environments
   - Concurrent access scenarios
2. **Ollama Framework** Fregly [2024], Gopal [2024], Linh [2024]
   - Individual workstations
   - Low-resource environments
   - Resource-constrained deployment

EHR Integration:

- RESTful API architecture (secure, authenticated endpoints)
- Real-time synchronization with patient records

Conclusion

Key Achievements:

- Built a vision-language model for automated endoscopic report generation, fine-tuned on a custom 32,545-sample Q&A dataset.
- Implemented a DPDP-compliant de-identification pipeline, making sensitive clinical data safe for AI training.
- Integrated system prompts and Chain-of-Thought reasoning to produce **concise, transparent, and clinically interpretable** reports.
- Achieved state-of-the-art: CIDEr 0.2667, ROUGE-L 0.4598, METEOR 0.4070, BLEU-4 0.3863, BERTScore 0.8216, report accuracy **84.03%**.

Impact: Drastically reduces documentation time; provides explainable AI assistance to gastroenterologists Eaton [2024], Hurst [2024]. **Path Forward:** Model quantization Wong [2024], SandLogic Technologies [2024] and clinical deployment trials (pending IEC approval) will bring the tool into real-world endoscopic workflows.

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